FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-064B - PharmaSpark Spark Gate

SEALED fresh full retest - explicit assignment only.

Artifact	C3-064B
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark | 16 lessons | 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 | Python 3.10+ | Java 17+ | learner local[2]
Brand	Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6 | AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

PharmaSpark scenario

Inventory movements join warehouse metadata. One quantity is invalid and W1 is intentionally hot. This is a fresh full retest, not a renamed MeterSpark solution.

Gate B/C remains sealed after a failed prior Gate and after remediation. It opens only when Mentor/reviewer explicitly assigns a fresh full retest.

Independent requirements

• Use explicit schemas and the laptop-safe local Spark profile.

• Create clean + rejected/flagged outputs; never silently delete suspect data.

• Prove dimension join-key uniqueness before joining.

• Produce grouped operational metrics at a stated output grain.

• Add one deterministic window metric that preserves fact-row grain.

• Write curated Parquet and justify partition/file strategy.

• Inspect a formatted plan and identify shuffle/join operators.

• Measure key distribution and classify hot-key/extreme behavior as skew risk, data-quality risk, or both.

• Demonstrate one evidence-based performance decision involving broadcast, repartition/coalesce, caching or AQE.

• Provide at least two automated transformation/invariant tests.

• Package a repeatable CLI-style job with version/run/output validation evidence.

• Explain one design change when moving from local[2] to a real cluster/Fabric Spark runtime.

Competency Gate - every competency required

Competency	PASS evidence
Correctness & requirements	Observed, validated, reproducible and explainable on this fresh case.
Data integrity & validation	Observed, validated, reproducible and explainable on this fresh case.
Reliability & rerun safety	Observed, validated, reproducible and explainable on this fresh case.
Testing & change safety	Observed, validated, reproducible and explainable on this fresh case.
Observability & diagnosis	Observed, validated, reproducible and explainable on this fresh case.
Security & evidence hygiene	Observed, validated, reproducible and explainable on this fresh case.
Maintainability & reproducibility	Observed, validated, reproducible and explainable on this fresh case.
Design reasoning & ownership	Observed, validated, reproducible and explainable on this fresh case.

PASS only if every required competency passes AND zero unresolved critical failures remain.

Evidence packet

Evidence	Your saved proof/path/note
Spark/Python/Java/session version proof	
Input/output grain statement	
Explicit schema evidence	
Clean/rejected reconciliation	
Dimension-key uniqueness + join row-count proof	
Grouped metric validation	
Window metric grain/tie rule	
Formatted plan annotation	
Skew/key-distribution evidence	
One controlled performance decision	
PySpark tests/invariants	
Parquet output/file evidence	
CLI/runbook/handoff	
Cluster/Fabric migration reasoning	